

Energy-Guided Priority under Battery Uncertainty for Multi-Robot Coordination

Hoseong Jung, Sungil Son, Inkyu Jang, H. Jin Kim, Jungwon Park*

Abstract—As multi-agent systems transition from structured environments to unpredictable domains, communication constraints and energy autonomy become critical operational bottlenecks. Furthermore, heterogeneous usage introduces battery uncertainty across identical hardware, complicating reliable planning. To address these challenges, we explicitly calibrate this usage-induced uncertainty to enable a highly scalable, low-bandwidth coordination strategy. We explore right-of-way coordination as a downstream task, where exchanging the energy sufficiency margin serves as a precedence cue during close encounters. Specifically, future energy needs are estimated via a cell-specific proxy model, with prediction errors rigorously calibrated using adaptive conformal inference. These calibrated margins dictate a unique yielding order, embedded into a responsibility-aware control barrier function (CBF) to enforce asymmetric collision avoidance. While validations show this precedence cue maintains safety with slight energy efficiency gains during local encounters, the calibrated framework is ultimately more suited for comprehensive mission planning to achieve true long-term autonomy in the wild.

I. INTRODUCTION

A. Motivation

Deploying multi-agent systems in unstructured environments represents a critical frontier for modern autonomous systems beyond structured workspaces [1], [2]. Unlike controlled settings that benefit from reliable communication infrastructure, field deployments are fundamentally limited by intermittent networks and bounded energy reserves [3], [4]. When agents cross paths in these shared environments, their coupled motions require a decentralized right-of-way decision to determine which agent should proceed and which should yield [5], [6]. In conventional approaches, resolving these spatial conflicts and preventing deadlocks requires bandwidth-intensive exchanges of complex planned trajectories or future task intents [7], [8]. To drastically minimize this communication overhead, we propose energy-guided priority as a highly scalable coordination strategy. By exchanging only the energy sufficiency margin that predicts the energy remaining after task completion, agents can establish a unique, pairwise yielding order without relying on heavy trajectory sharing [9], [10]. This metric naturally scales to field constraints like sparse charging and complex missions, serving as a robust yielding cue for lifelong autonomy.

Hoseong Jung, Sungil Son, and H. Jin Kim are with Seoul National University, South Korea {ghtjdaleka, aldridge99, hjinkim}@snu.ac.kr.

Inkyu Jang is with the University of California, Berkeley, USA janginkyu.larr@gmail.com.

Jungwon Park is with Seoul National University of Science and Technology, South Korea jungwonpark@seoultech.ac.kr.

*Corresponding author

While this energy-guided priority is communication-efficient, its practical application must account for battery dynamics over long-term operations. In practice, heterogeneous usage profiles naturally cause cell-specific battery degradation, introducing uncertainty even among identical hardware specifications. Rather than treating this uncertainty as a critical failure point, we seek to explicitly calibrate it to maintain a consistent and reliable precedence cue.

In this work, we estimate future energy requirements via a cell-specific proxy model and bound the residual prediction errors using adaptive conformal inference [11]. The resulting calibrated margins determine the discrete yielding order. We then embed this precedence into a responsibility-aware control barrier function (CBF) formulation [12]. Fig. 1 illustrates the overall idea: calibrated energy margins establish the yielding order, which is subsequently reflected in collision-avoidance responsibility.

B. Summary of Contributions

We develop an uncertainty-calibrated energy-guided precedence rule for multi-robot right-of-way coordination and implement it using a responsibility-aware CBF formulation. Our main contributions are summarized as follows.

Energy Sufficiency as a Precedence Criterion. We use the energy expected to remain after task completion and travel to a charging station to determine pairwise right-of-way. By exchanging only this scalar, agents establish a unique yielding order based on energy conditions, reducing communication overhead for conflict resolution.

Uncertainty-Calibrated Energy Estimates. We estimate the energy required to reach a charging station directly and after completing the assigned task using a battery-specific proxy model. Adaptive conformal inference converts these predictions into bounds whose long-run violation rate is controlled at a prescribed level. The resulting bounds define conservative precedence margins and an energy-sufficiency certificate with a conditional station-reachability guarantee.

Safety-Preserving Precedence Integration. We encode the precedence in a responsibility-aware CBF formulation by assigning unequal avoidance contributions within each interacting pair. The paired local constraints preserve the underlying collision-avoidance condition regardless of which robot yields. When multiple local right-of-way conflicts arise, all pairwise decisions follow a common energy-based ordering, thereby preventing cyclic precedence.

C. Limitations and Future Works

Our evaluations validate the efficacy of the energy-guided precedence cue within localized multi-agent encounters.

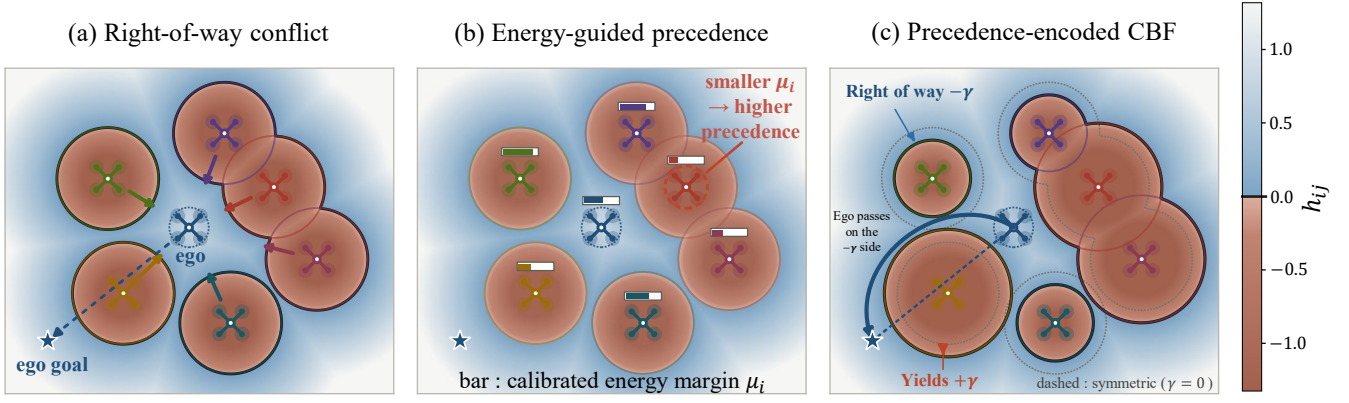


Fig. 1: Energy-guided right-of-way coordination. (a) Multiple robots compete for right of way in a congested encounter. (b) Calibrated precedence cues μ_i determine a unique yielding order, where smaller μ_i indicates higher precedence and larger μ_i indicates greater flexibility to yield. (c) The order is encoded through a responsibility-aware CBF term (γ), assigning greater avoidance responsibility to the yielding robot while preserving pairwise safety.

However, confining this calibrated margin exclusively to a CBF formulation is inherently myopic for system-wide energy optimization. To achieve lifelong energy autonomy in disaster scenarios, future work will integrate this framework into higher-level mission planning architectures, such as global task allocation and long-horizon pathfinding.

II. PROBLEM FORMULATION

This section presents the multi-robot right-of-way problem under uncertain battery conditions. We first define the robot motion, battery state, and future energy requirements for task completion and station reachability (Sec. II-A). We then formulate the energy-guided precedence and safety objectives (Sec. II-B).

A. Motion and Energy Dynamics

Consider a team of N robots, indexed by $\mathcal{A} = \{1, \dots, N\}$. Robot i follows the control-affine dynamics

$$\dot{x}_i = f_i(x_i) + g_i(x_i)u_i, \quad u_i \in \mathcal{U}_i \quad (1)$$

where $x_i \in \mathcal{X}_i$ is the robot state, $\mathcal{U}_i$ is a convex input set, and f_i and g_i are locally Lipschitz. Let $p_i = \Pi_i x_i$ denote the robot position. Each robot is assigned a task goal p_i^g and may recharge at one of the stations $\mathcal{C} = \{c_1, \dots, c_K\}$.

We represent the battery state of robot i by its filtered load voltage V_i , which serves as a measurable proxy for remaining energy. Let $V_{\min}$ denote the minimum operational voltage, and define the augmented state

$$z_i := [x_i^\top, V_i]^\top. \quad (2)$$

For each station c_k , let $c_{i,k}^{\text{sta}}(z_i)$ denote the future voltage cost of traveling directly to c_k , and let $c_{i,k}^{\text{task}}(z_i)$ denote the cost of completing the assigned task and subsequently reaching c_k . These trajectory-level costs depend on the current motion and battery states and are unknown at the time of coordination.

B. Problem Statement

For multi-robot right-of-way coordination, we aim to 1) construct uncertainty-calibrated upper bounds for future energy requirements; 2) use these bounds to determine a

unique energy-guided yielding order; and 3) encode the resulting order via responsibility-aware CBF constraints while preserving pairwise collision avoidance. The problem is formulated as follows.

Problem 1 (Energy-Guided Right-of-Way Coordination). *Given the robot dynamics in Eq. (1), the measured augmented states $\{z_i\}_{i \in \mathcal{A}}$, and a target violation level $\delta \in (0, 1)$, construct calibrated future-cost bounds and control inputs satisfying the following requirements.*

- i) **Uncertainty-calibrated energy bounds:** for each calibrated sequence of future energy requirements $\{c_\ell\}$, construct upper estimates $\{\bar{c}_\ell\}$ such that

$$\limsup_{L \rightarrow \infty} \frac{1}{L} \sum_{\ell=1}^L \mathbb{1}\{c_\ell > \bar{c}_\ell\} \leq \delta. \quad (3)$$

- ii) **Energy-guided precedence:** Using the uncertainty-calibrated future energy estimates, construct energy-based precedence scores $\{\mu_i\}_{i \in \mathcal{A}}$ that reflect each robot's flexibility to accommodate the additional cost of yielding. For every interacting pair (i, j) , the robot with the larger score is assigned to yield, with ties resolved by a fixed rule. When multiple conflicts arise, all pairwise decisions follow the common ordering induced by $\{\mu_i\}_{i \in \mathcal{A}}$.
- iii) **Safety-preserving precedence integration:** encode the selected yielding order through complementary responsibility allocations while ensuring

$$h_{ij}(x_i(0), x_j(0)) \geq 0 \implies h_{ij}(x_i(t), x_j(t)) \geq 0, \quad (4)$$

for all $t \geq 0$ and every interacting pair (i, j) .

III. METHOD

This section solves Problem 1 in two stages: uncertainty-calibrated energy-based precedence cue (Sec. III-A), and safety-preserving precedence integration (Sec. III-B). The overall procedure of our approach is summarized in Fig. 2.

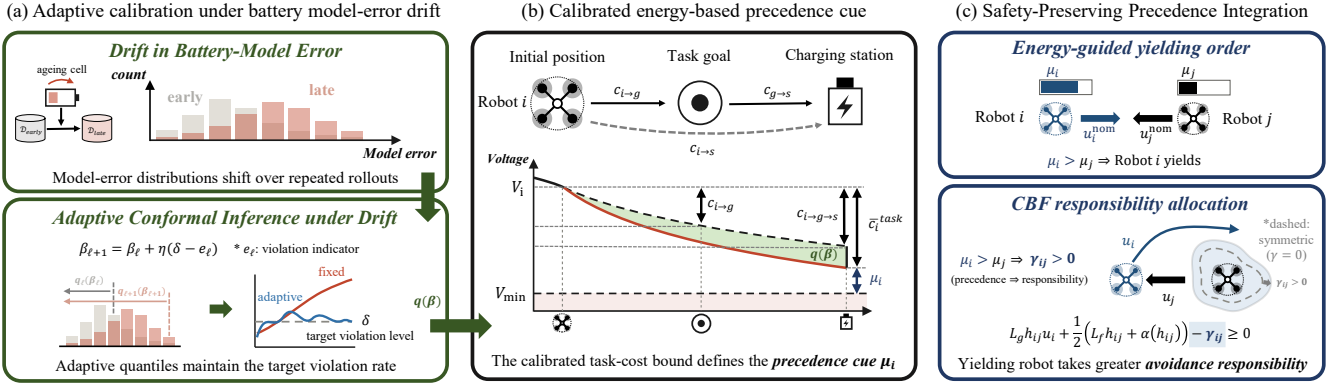


Fig. 2: Overview of the proposed energy-guided precedence framework. (a) Adaptive conformal inference calibrates battery-model error under distribution shift, producing an adaptive quantile $q_\ell(\beta_\ell)$ at the target violation level. (b) The calibrated task-cost bound defines the energy-based precedence cue μ_i . (c) The resulting yielding order is encoded through CBF responsibility allocations, so the yielding robot assumes greater avoidance responsibility while pairwise safety is preserved.

A. Uncertainty-Calibrated Precedence Cue

To establish a low-bandwidth yielding order, we define the energy sufficiency margin μ_i as the calibrated estimate of remaining energy after task completion. Let c_l denote the true future energy cost and $\hat{c}_l$ be the prediction from a cell-specific proxy model. To handle usage-induced battery uncertainty, we calibrate the residual error $s_l = c_l - \hat{c}_l$ using adaptive conformal inference [11]. Specifically, the adaptive quantile $q_l(\beta_l)$ is computed as the empirical $(1 - \beta_l)$ -quantile of the previously observed calibration scores. This feedback mechanism dynamically adjusts the effective misscoverage level: a bound violation decreases β to enlarge subsequent prediction bounds, whereas repeated coverage permits tighter margins. Consequently, the online update maintains a target violation rate δ even under distribution shifts caused by cell degradation:

$$\beta_{l+1} = \beta_l + \eta(\delta - \mathbb{1}\{c_l > \hat{c}_l + q_l(\beta_l)\}). \quad (5)$$

The resulting calibrated task-cost bound $\bar{c}_l^{\text{task}}$ defines the precedence margin:

$$\mu_i = V_i - V_{\min} - \bar{c}_l^{\text{task}}, \quad (6)$$

where V_i is the current voltage and $V_{\min}$ is the minimum operational threshold. For any interacting pair (i, j) , the agent with the larger margin has greater flexibility and is assigned to yield (i.e., $\mu_i > \mu_j \implies i$ yields to j).

B. Safety-Preserving Precedence Integration

To enforce this discrete precedence order at the continuous control level, we embed it into a responsibility-aware control barrier function (CBF) [12]. We define a directional indicator $\psi_{ij} \in \{-1, 1\}$ based on the yielding order and allocate an asymmetric avoidance responsibility $\gamma_{ij} = \rho\psi_{ij}$. The yielding agent takes on a tighter constraint within the local quadratic program (QP):

$$L_{g_i} h_{ij} u_i + \frac{1}{2} (L_{f_{ij}} h_{ij} + \alpha(h_{ij})) - \gamma_{ij} \geq 0. \quad (7)$$

By exchanging only the scalar μ_i , agents dynamically reallocate avoidance efforts without sharing future trajectories, strictly preserving pairwise collision avoidance.

IV. EXPERIMENTS

This section validates the proposed framework in two stages: real-world battery calibration (Sec. IV-A), and hardware validation of energy-guided precedence (Sec. IV-B).

A. Real-World Battery Calibration

Setup and Dataset. To capture realistic discharge statistics under dynamic flight profiles, we collected an extensive dataset using single-cell LiPo batteries from a Crazyflie 2.1 fleet. Rather than relying on static bench tests, we compiled approximately 3 hours of active flight data across 200 distinct sorties, incorporating both standard demonstrations [13] and multi-sortie waypoint navigation tasks. Crucially, the data was sourced from 14 individual battery cells, explicitly capturing the heterogeneous voltage drop characteristics encountered in physical deployments. We fit a physically parameterized load-voltage proxy model to this dataset, evaluating the calibration performance using a strict hold-out testing approach for each independent cell.

Results. We evaluate whether adaptive conformal inference reliably maintains the one-sided energy bounds below the target violation level ($\delta = 0.10$). As illustrated in Fig. 3 (left), the adaptive framework successfully drives the empirical violation frequency below the target threshold over sequential calibration rounds. Furthermore, when evaluated across all 14 individual battery cells under distribution shift, our adaptive method robustly bounds the error (Fig. 3 (right)). In contrast, a standard fixed-quantile baseline fails to adapt to cell-specific degradation, resulting in uncalibrated bound violations with a mean of 0.116 and a standard deviation of 0.114. These comparative results confirm that our calibration layer effectively absorbs usage-induced discrepancies, producing a statistically sound energy-sufficiency margin to serve as a trustworthy precedence cue.

B. Hardware Validation of Energy-Guided Precedence

Setup. We deployed eight Crazyflie 2.1 quadrotors under an OptiTrack motion capture system to execute a dense position-exchange task, utilizing Crazyswarm to broadcast nominal trajectories [14]. Given the close proximity of

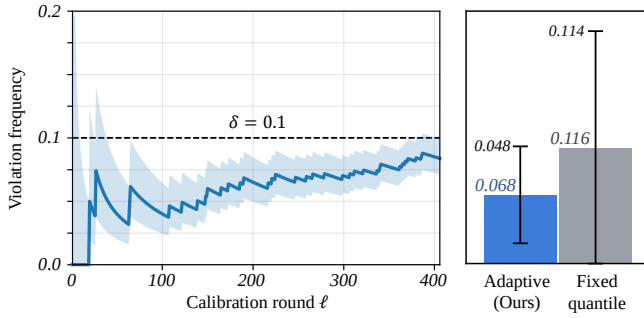


Fig. 3: Battery calibration performance. **(Left)** The running violation frequency converges below the target $\delta = 0.1$. **(Right)** Our adaptive method successfully bounds errors across 14 cells (mean 0.068, std 0.048), outperforming the fixed-quantile baseline.

the congested intersection, agents reliably exchanged their standard local states (e.g., pose and velocity) alongside their individual calibrated energy margins. While the local state information was used to enforce a responsibility-aware CBF with a 0.4 m safety margin, the single scalar energy margin uniquely determined the pairwise yielding precedence. This critical separation allows agents to safely coordinate without relying on bandwidth-intensive trajectory sharing.

Results and Implications. Throughout the demonstration, this decentralized approach successfully resolved all spatial conflicts, resulting in zero collisions. As illustrated in Fig. 4, the precedence cue induced a strictly asymmetric yielding behavior: agents with lower energy margins (depicted in red) retained their right-of-way for direct traversal, whereas agents with higher margins (depicted in yellow) actively absorbed the collision-avoidance responsibility to yield safely. Ultimately, this experiment confirms that coupling safety-critical state feedback with a low-overhead, energy-guided priority signal provides a scalable, robust foundation for long-term multi-agent operations in the field.

V. CONCLUSION

This work introduced an uncertainty-calibrated, energy-guided precedence rule for multi-agent right-of-way coordination. To minimize communication overhead, we absorbed battery degradation via adaptive conformal inference and embedded this reliable cue into a control barrier function. While hardware validations demonstrated its efficacy for decentralized collision avoidance, confining this metric exclusively to the reactive control layer remains myopic for system-wide optimization. To achieve true energy autonomy, future work will integrate this framework into lifelong multi-agent reinforcement learning architectures for global task allocation in disaster scenarios.

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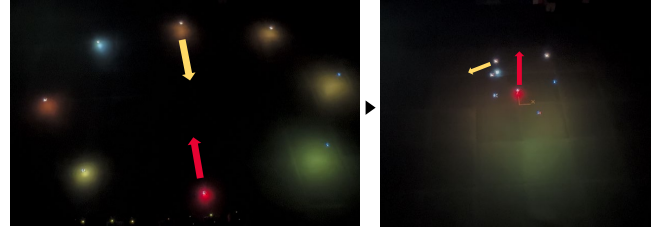


Fig. 4: Hardware validation in an 8-agent congested intersection. **(Left)** Agents move to opposite positions, creating spatial conflicts. The red agent possesses a higher energy-guided priority than the yellow agent. **(Right)** Enforcing the calibrated precedence cue, the yellow agent actively yields, allowing the red agent to safely maintain its direct path.

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